

Mathematics XI

Time Allowed : 1 Hour _____ **Maximum Marks : 60**

Please read the instructions carefully. You will be allotted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. This question paper consists of 6 questions carrying 10 marks each.

Q1: Evaluate $\sum_{k=1}^{11} (2 + 3^k)$

Q2: If a, b, c and d are in G.P. show that

$$(a^2 + b^2 + c^2)(b^2 + c^2 + d^2) = (ab + bc + cd)^2$$

Q3: If A and G be A.M. and G.M., respectively between two positive numbers, prove that the numbers are $A \pm \sqrt{(A+G)(A-G)}$

Q4: Find the value of n so that $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ may be the geometric mean between a and b.

Q5: Find the sum of the sequence 7, 77, 777, 7777, ... to n terms.

Q6: The difference between any two consecutive interior angles of a polygon is 5° . If the smallest angle is 120° , find the number of the sides of the polygon.